

Class 11 Biology: High-Yield Diagram Checklist

Chapter 3: Plant Kingdom (NCERT Page-wise Priority & Labelling Guide)

Instructions for Students: Practice drawing each diagram clearly with pencil and write all labels in neat BLOCK letters. Pay special attention to distinguishing male vs. female thalli and life-cycle phases (sporophyte vs. gametophyte).

Fig No.	Diagram / Organism	NCERT	Priority	Key Labelling Points / Identification
Fig 3.2 (a, b)	<i>Marchantia</i> (Liverwort) Female & Male Thalli	Pg 28	★★★ Top Priority	Archegoniophore (female), Antheridiophore (male), Gemma cup, Thallus, Rhizoids
Fig 3.2 (c)	<i>Funaria</i> (Moss) Gametophyte & Sporophyte	Pg 28	★★★ Top Priority	Sporophyte: Capsule, Seta, Foot. Gametophyte: Leaves, Main axis, Rhizoids
Fig 3.1 (b)	Brown Algae (Phaeophyceae) <i>Laminaria</i> , <i>Fucus</i> , <i>Dictyota</i>	Pg 25	★★★ Top Priority	Frond (leaf-like), Stipe (stalk), Holdfast (attachment), Air bladder, Midrib (in <i>Fucus</i>)
Fig 3.3 (b)	<i>Equisetum</i> (Horsetail)	Pg 31	★★★ Top Priority	Strobilus (cone), Node, Internode, Branches, Underground Rhizome
Fig 3.3 (a, d)	<i>Selaginella</i> & <i>Salvinia</i> (Pteridophytes)	Pg 31	★★☆ High Priority	<i>Selaginella</i> : Stem, Leaves (microphylls), Roots, Strobilus. <i>Salvinia</i> : Floating aquatic fronds, roots
Fig 3.1 (a, c)	Green & Red Algae <i>Volvox</i> , <i>Ulothrix</i> , <i>Porphyra</i> , <i>Polysiphonia</i>	Pg 25	★★☆ High Priority	<i>Volvox</i> : Daughter colony, Parent colony. <i>Polysiphonia</i> : Main axis, Branches. <i>Porphyra</i> : Frond
Fig 3.2 (d)	<i>Sphagnum</i> (Peat Moss)	Pg 28	★★☆ High Priority	Antheridial branches, Archegonial branches, Branch clusters
Fig 3.4 (c)	<i>Ginkgo</i> (Gymnosperm)	Pg 33	★★☆ High Priority	Long shoot, Dwarf shoot, Fan-shaped leaves, Seeds (naked)
Fig 3.4 (a, b) & 3.5	<i>Cycas</i> , <i>Pinus</i> & Angiosperms	Pg 33, 34	★★☆ Moderate	Pinnate leaves (<i>Cycas</i>), Needle foliage (<i>Pinus</i>), Dicot vs Monocot habit

KEY EXAM PREPARATION & DRAWING TIPS

- **Liverwort Distinction:** Always practice differentiating *Marchantia* female thallus (umbrella/star-like rays on archegoniophore) from male thallus (disc-shaped antheridiophore cap).
- **Gametophyte vs. Sporophyte:** In *Funaria*, clearly mark the boundary where the dependent diploid sporophyte (Foot, Seta, Capsule) attaches to the leafy haploid gametophyte.
- **Table 3.1 Division Summary:** Alongside diagrams, memorize Table 3.1 (Pg 27) for pigments, stored food (Floridean starch, Mannitol/Laminarin), and flagellar insertion.